MANCHESTER 1824
The University of Manchester

EART11100 Understanding the Earth

Week 1 Clocks in Rocks: Time and the Stratigraphic Record

The aims of this week's workshop are:

- to explore how geologists reconstruct geologic history using stratigraphic methods

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Understanding the Earth

Week 1 Clocks in Rocks: Time and the Stratigraphic Record

On successful completion of the workshop you will be able to:

- apply stratigraphic principles to infer whether one rock is older than another
- classify unconformities according to the relationships between the layers above and below them
- write a sequence or geologic events

Principle of original horizontality and superposition



Sediments usually form flat-lying deposits on the Earth's surface

Examples of exceptions





Sediments are only deposited on steep slopes over short distances

Principle of original horizontality and superposition



Sediments usually form flat-lying deposits on the Earth's surface



Undeformed sedimentary rocks are horizontal because the original sediments were horizontal

youngest

oldest

Principle of cross-cutting relations



A geologic feature (e.g., an igneous rock unit or fault) that cuts across another rock unit must be younger than the unit it cuts across.

Folding/faulting/metamorphism



A word of caution!



C

B

A

Principle of inclusion: Example 1



A rock unit that contains pieces of pre-existing rocks (inclusions) must be younger than the rock unit from which the inclusions came

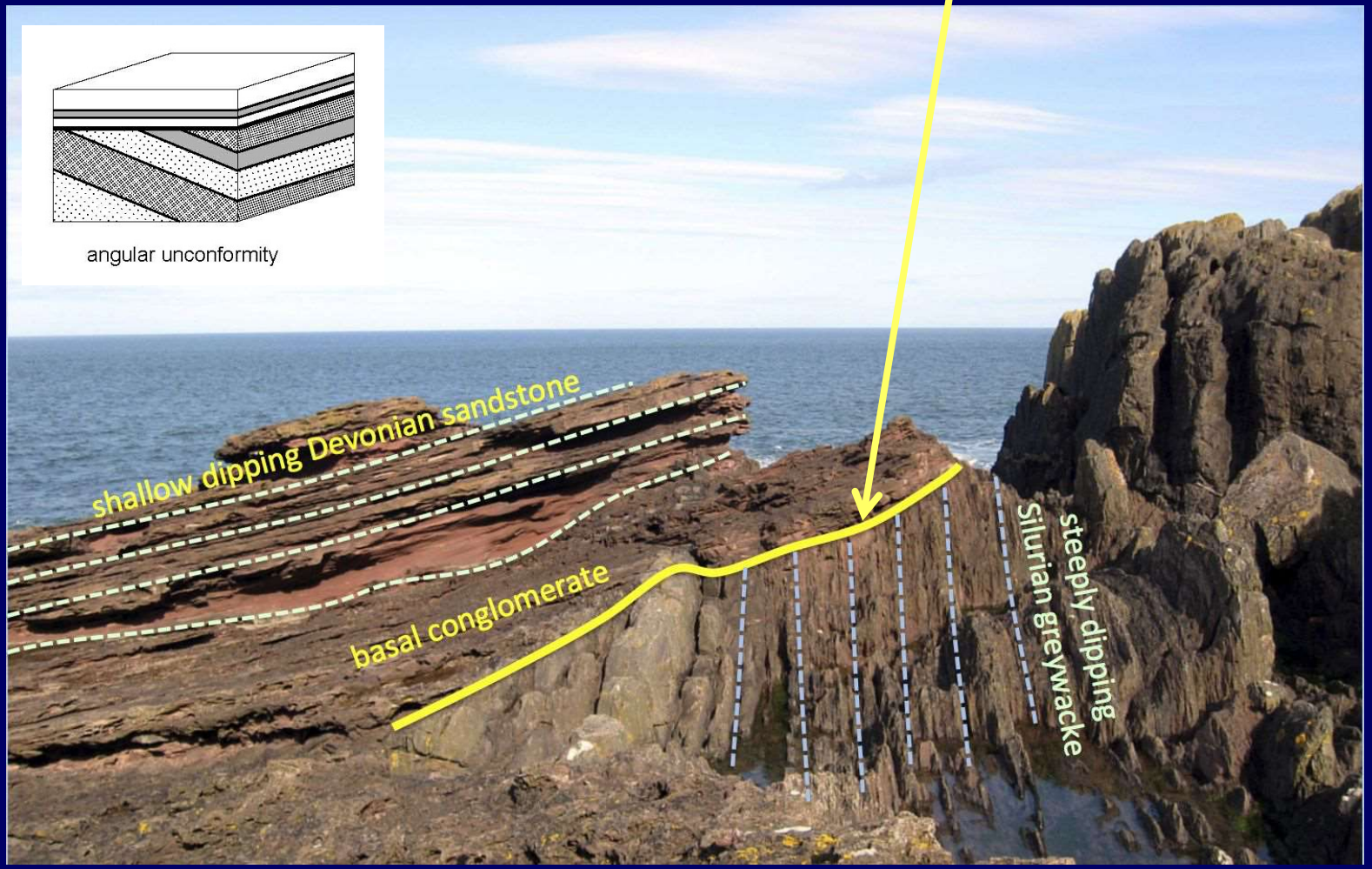
Principle of inclusion: Example 2





Hiatus in deposition- a missing record of time!

Hiatus: erosion or non-deposition
Unconformity: regional hiatus



Siccar Point, Scotland (James Hutton's angular unconformity)

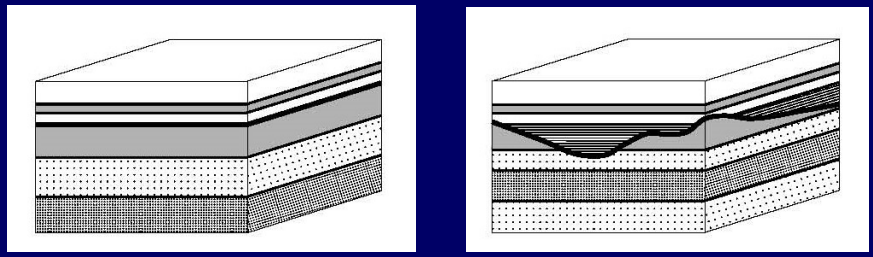
Unconformities: Nonconformity



Nonconformity between weathered Pikes Peak granite and overlying Paleozoic sandstone near Manitow Springs, Colorado

Unconformities: Disconformity





disconformity



Amroth, Pembrokeshire

The principles of stratigraphy: summary

- The principles of stratigraphy allow sequences of geological events to be established

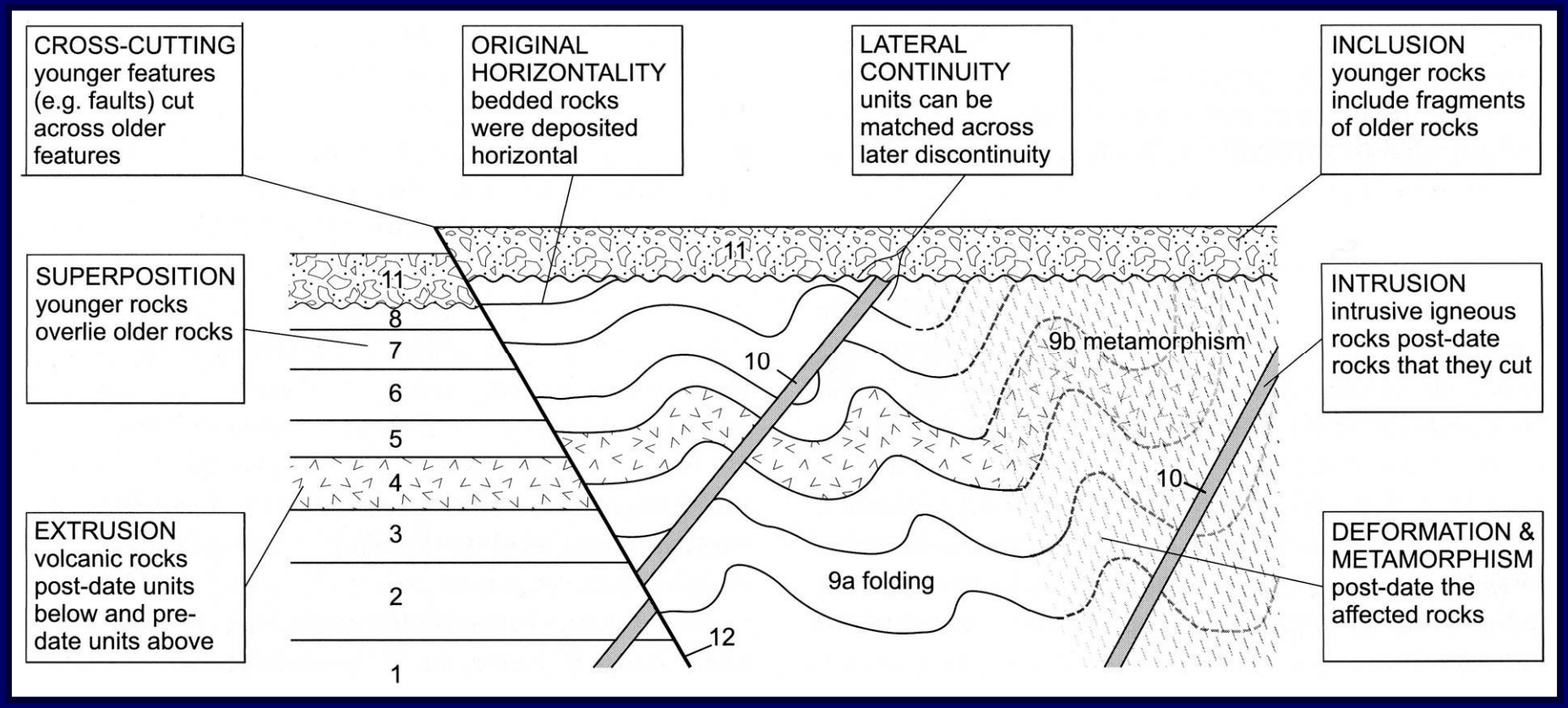


Illustration of the main principles of stratigraphy. The numbers indicate the sequence of events (1 = first → 12 = most recent). To be complete one might add, after 10 but before 11, uplift and erosion leading to the unconformity that is represented by the wavy line at the base of 11

Can you do what we set out to do?

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Suggested reading
Grotzinger and Jordan, 2020, Understanding Earth, 8th Edition, Chapter 9